



UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Lahore Campus

Design and Implementation of an Enterprise Campus Network with Zero Trust Security Architecture

Project Proposal

Submitted in Partial Fulfillment of the Requirements for the Course of Computer Networks

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ABSTRACT

This document presents a formal proposal for the design and simulation of an enterprise-grade campus network for the University of Management and Technology (UMT), Lahore. The network architecture is grounded in the Zero Trust Security model as defined by the National Institute of Standards and Technology in Special Publication 800-207, which prescribes that no user, device, or system process shall be granted implicit trust based solely on network location. Every access request must be explicitly verified, authorised, and continuously evaluated against defined policy.

The proposed design employs a three-tier hierarchical model comprising Core, Distribution, and Access layers, with five VLAN-based logical segments mapped directly to the physical buildings of the campus. Inter-VLAN communication is governed by Layer 3 switching and Access Control Lists applied at the distribution tier to prevent unauthorised lateral movement. Open Shortest Path First routing provides dynamic, convergent inter-segment reachability. DHCP automates IP address allocation, and Network Address Translation simulates controlled internet access for permitted user categories.

The full architecture is implemented and validated within Cisco Packet Tracer using a representative device set capable of logically scaling to a user population exceeding two thousand. The document covers requirement analysis, physical-to-logical building mapping, IP addressing, configuration methodology, testing procedures, and expected deliverables.

Keywords: Zero Trust Security, VLAN Segmentation, Hierarchical Network Design, OSPF, Extended ACL, Inter-VLAN Routing, Enterprise Network, Cisco Packet Tracer.

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1. INTRODUCTION AND BACKGROUND

Contemporary academic institutions operate complex, multi-stakeholder network environments in which heterogeneous devices, ranging from student laptops and laboratory workstations to administrative terminals and IP infrastructure appliances, share a common physical medium. This convergence of endpoints with distinct security classifications, access requirements, and behavioural profiles creates inherent risk when the underlying network provides no mechanism for enforcing access boundaries between organisational units.

The perimeter trust model, which grants implicit network access to all authenticated devices within a defined boundary, is no longer a viable security posture. The proliferation of Bring Your Own Device policies, the lateral reachability afforded by flat network topologies, and the sophistication of credential-based and insider threats collectively render perimeter-based defences inadequate. A single compromised endpoint within such an architecture provides an adversary unrestricted access to all co-resident systems.

The Zero Trust Security model, formalised by Forrester Research and codified by NIST in Special Publication 800-207, addresses these deficiencies by mandating that every access request be authenticated and authorised regardless of the requester's network location. Trust is never implied; it is earned through continuous verification against contextual policy. This model requires explicit network segmentation, least-privilege access enforcement, and comprehensive control of inter-segment traffic flows.

This proposal describes the design and simulation of a Zero Trust-compliant enterprise campus network for UMT Lahore. The architecture maps logically to the physical campus, assigns dedicated VLANs to distinct organisational units, and enforces inter-segment policy through Access Control Lists and Layer 3 routing. The simulation is executed in Cisco Packet Tracer to provide a controlled, reproducible environment for design validation.

2. PROBLEM STATEMENT

The network infrastructure representative of a typical academic institution of UMT's scale presents several compounding deficiencies that collectively undermine operational efficiency and information security. Each deficiency is examined in turn.

2.1 Absence of Network Segmentation

Without VLAN-based logical segmentation, all connected devices reside within a shared broadcast domain regardless of organisational role. Under this configuration, a compromised student device can transmit frames that reach faculty servers, administrative databases, and IT management infrastructure. The absence of Layer 2 boundaries enables broadcast amplification attacks, ARP spoofing, and unrestricted lateral movement by any device on the network.

2.2 Unrestricted Inter-Departmental Access

Flat network architectures provide no enforcement point at which role-based traffic restrictions can be applied. A student workstation has the same Layer 3 reachability as an administrative terminal, violating the principle of least-privilege access. Without policy enforcement mechanisms such as Extended ACLs applied at a routing boundary, controlling this traffic is operationally infeasible at scale.

2.3 Static IP Address Management

Manual IP address assignment across a large campus environment is error-prone, administratively intensive, and difficult to audit. Address conflicts, incorrect default gateway assignments, and untracked device inventories are common consequences. These conditions cause intermittent connectivity failures that disproportionately affect critical institutional services.

2.4 Implicit Trust Architecture

The prevailing implicit trust model confers broad internal access rights upon successful network authentication. This posture is exploited by ransomware propagation, data exfiltration by malicious insiders, and credential-based attacks. The Zero Trust framework corrects this by requiring identity and context-aware authorisation at every access request, regardless of the requester's network membership.

2.5 Scalability Constraints

Unstructured network expansions driven by increasing student enrolment and proliferating endpoint devices, executed without a coherent hierarchical design, result in routing inefficiencies, Spanning Tree

instability, and management complexity that impedes institutional growth. A structured hierarchical model is prerequisite to controlled, predictable scaling.



3. OBJECTIVES

The following objectives define the measurable success criteria for this project. Each objective is stated in terms of a verifiable outcome.

1. Design and implement a three-tier hierarchical campus network comprising Core, Distribution, and Access layers in Cisco Packet Tracer. The architecture must logically support a minimum of two thousand concurrent users using a representative simulation device set.
2. Deploy and configure five discrete VLANs corresponding to Student, Faculty, Administration, IT Department, and Guest user categories. Each VLAN must be mapped to a specific physical campus building with complete Layer 2 isolation between segments confirmed by test.
3. Implement OSPF dynamic routing and Inter-VLAN routing via Layer 3 Switched Virtual Interfaces, achieving verified end-to-end connectivity within authorised segments and confirmed denial of unauthorised inter-VLAN traffic in accordance with defined ACL policy.
4. Establish role-based access control through Extended Named ACLs that restrict Student and Guest VLANs from initiating connections to Administration and IT Department segments, while permitting Faculty access to designated shared resources under defined conditions.
5. Validate Zero Trust compliance through systematic connectivity testing, ACL rule verification, and OSPF convergence testing, producing documented test reports that confirm policy enforcement across all VLAN boundaries against each stated objective.

4. SCOPE OF THE PROJECT

4.1 Included Within Scope

- Three-tier hierarchical network design implemented and validated in Cisco Packet Tracer 8.x
- VLAN creation and 802.1Q trunk link configuration across all inter-switch connections with selective VLAN pruning
- Inter-VLAN routing via Switched Virtual Interfaces on Layer 3 distribution switches
- OSPF Area 0 dynamic routing configured on all Layer 3 devices
- Centralised DHCP with per-VLAN scopes and relay agent configuration on distribution SVIs
- Simulated NAT with PAT on the internet gateway router representing controlled WAN access
- Extended Named ACL design, implementation, and validation for role-based traffic control
- Physical campus building to VLAN logical mapping
- Comprehensive test plan and documented results verifying connectivity and ACL enforcement
- Port security, DHCP snooping, and STP hardening as post-validation optimisation measures
- Wireless access point integration as a supplementary enhancement for Student, Faculty, and Guest segments

4.2 Excluded From Scope

- Physical hardware procurement, rack installation, structured cabling, or site survey
- Live integration with identity providers including RADIUS, LDAP, or Microsoft Active Directory
- Software-Defined Networking, OpenFlow, or network virtualisation beyond Packet Tracer capabilities
- Real-time performance benchmarking or traffic load testing under production-scale conditions
- Cloud integration, hybrid WAN, or SD-WAN architecture
- Intrusion Detection and Prevention Systems
- Public Key Infrastructure or certificate-based mutual authentication
- IPv6 addressing or OSPFv3 configuration

5. METHODOLOGY

The project follows a structured, phase-driven methodology aligned with the Cisco Hierarchical Network Design model and the NIST SP 800-207 Zero Trust Architecture framework. Each phase produces verifiable outputs that contribute directly to the final deliverable. Phases are executed sequentially, with each phase validated before the subsequent phase commences.

5.1 Requirement Analysis

Requirement analysis establishes the design constraints, user population parameters, and security policy baseline governing all subsequent design decisions. The network serves five distinct user categories whose access requirements and security classifications are fundamentally different.

5.1.1 User Population and Traffic Characterisation

User Category	Approx. Count	Primary Access Requirements	Security Classification
Students	1,400	Internet, LMS, Student Portal	Low privilege
Faculty	200	Research Servers, Admin Portal, Internet	Medium privilege
Administration	100	ERP, HR Systems, Restricted Databases	High privilege
IT Department	50	All segments, Network Management	Highest privilege
Guests	250	Internet access only	Isolated, no internal reachability

5.1.2 Zero Trust Security Policy Baseline

The security policy baseline is derived from the principle of least-privilege access and defines the non-negotiable access rules that govern all inter-VLAN communication. These rules precede network design and drive the ACL configuration in Section 5.5.6.

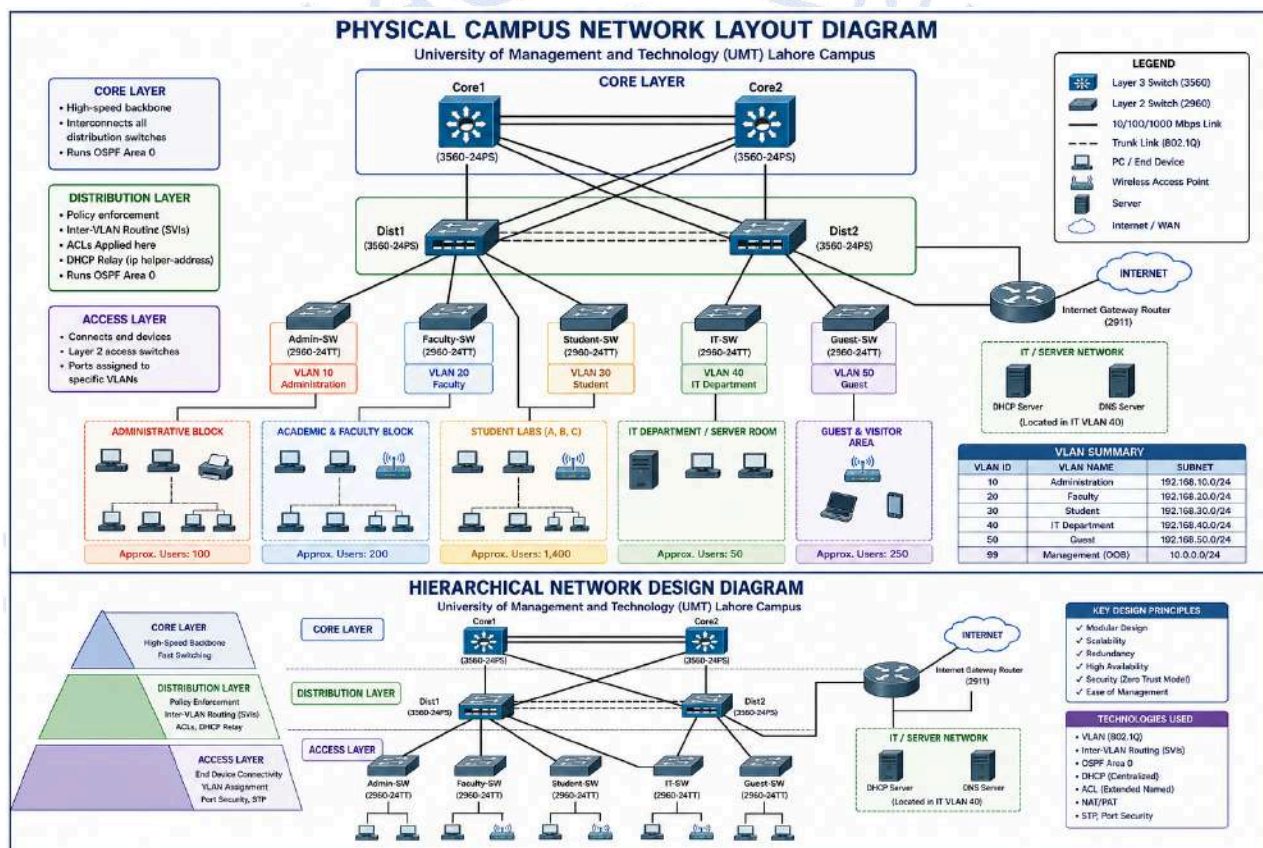
- No VLAN shall have unrestricted access to any other VLAN by default. All inter-VLAN communication must be explicitly permitted by an ACL rule.
- Guest VLAN traffic is confined to NAT-translated internet access only. All inbound connections from the Guest VLAN to any internal segment are denied.
- Student VLAN devices must not initiate connections to Administration or IT Department networks under any circumstances.

- Faculty VLAN may access designated shared administrative resources but is denied access to IT infrastructure management interfaces.
- IT Department VLAN retains management access to all segments exclusively for network administration purposes, subject to explicit ACL permit statements.



5.2 Network Design and Physical Campus Mapping

The network architecture is structured according to the Cisco three-tier hierarchical model. This model separates network functions into distinct operational tiers to optimise forwarding performance, enforce policy at designated boundaries, and provide a framework for controlled scalability.



5.2.1 Physical Campus Building to VLAN Mapping

Each physical building on the UMT campus is assigned a dedicated VLAN, establishing a direct correspondence between physical location and logical network segment. This alignment simplifies fault isolation, enforces physical access boundaries, and treats each building as a distinct trust zone consistent with the Zero Trust boundary model.

Building	VLAN Name	VLAN ID	Layer	Representative Devices
Administrative Block	Administration	VLAN 10	Access	Workstations, Printers
Academic and Faculty Block	Faculty	VLAN 20	Access	Workstations, Access Points

Student Laboratories A, B, C	Student	VLAN 30	Access	Lab PCs, Access Points
IT Department and Server Room	IT Department	VLAN 40	Access and Distribution	Servers, Management PCs
Guest and Visitor Area	Guest	VLAN 50	Access (Isolated)	Access Points, Guest Terminals

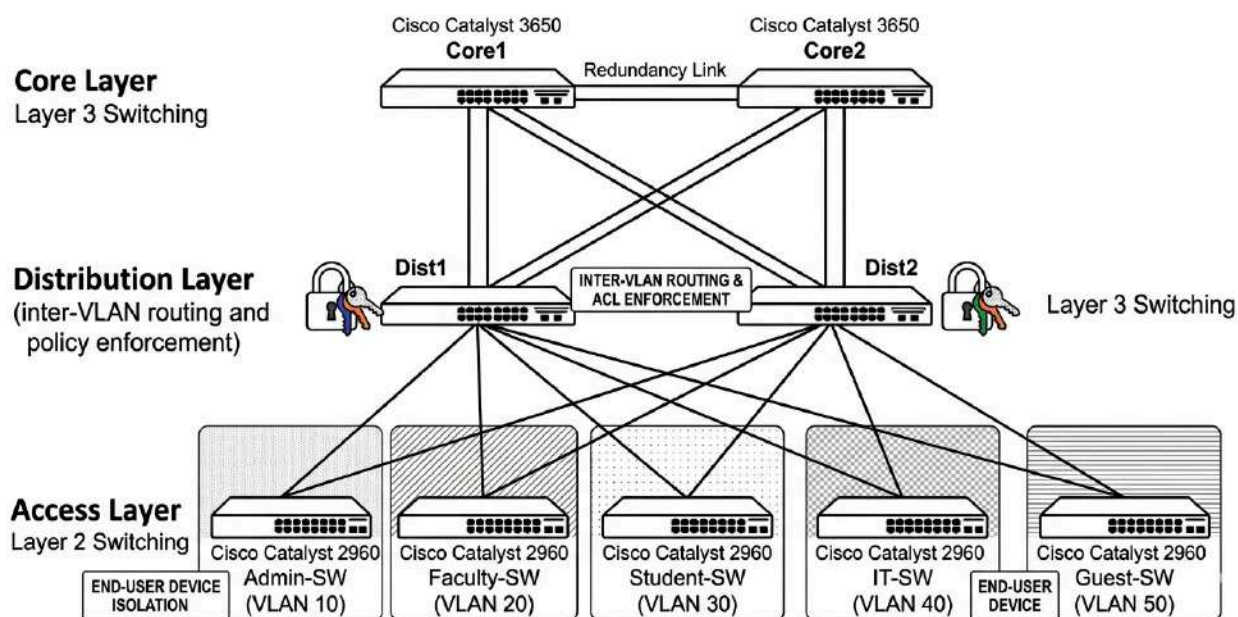
5.2.2 Hierarchical Layer Definitions

Core Layer. Two Layer 3 core switches form the network backbone, providing high-capacity, low-latency interconnection between distribution tier switches. The core layer performs no policy enforcement and executes no ACL inspection. Its sole function is rapid packet forwarding. OSPF adjacency is established at this tier to ensure loop-free dynamic routing.

Distribution Layer. Two Layer 3 distribution switches serve as the primary policy enforcement boundary within the Zero Trust architecture. These devices host Switched Virtual Interfaces for each VLAN, perform Inter-VLAN routing, and apply Extended ACLs on inbound SVI interfaces to regulate inter-segment traffic. All trunk uplinks from access layer switches terminate at this tier. The distribution layer aggregates user traffic and forwards it to the core based on OSPF-computed routing tables.

Access Layer. One Layer 2 switch is deployed per campus building to connect end-user devices to the network. Access ports are configured as untagged members of a single VLAN. Uplink ports to distribution switches are configured as 802.1Q trunk links carrying all VLANs active in that building. End devices have no visibility of VLAN tagging operations.

UMT Lahore Campus Enterprise Network – Zero Trust Architecture



5.3 IP Addressing Scheme

The IP addressing scheme employs RFC 1918 private address space structured into /24 subnets, each providing 254 usable host addresses. This allocation is deliberately over-provisioned relative to the simulated device count to demonstrate logical scalability to the target user population without requiring address space reorganisation. A dedicated Management VLAN provides out-of-band access to all network infrastructure devices using statically assigned addresses, ensuring that management traffic is isolated from user data planes.

VLAN Name	ID	Subnet	Default Gateway	DHCP Range	Broadcast
Administration	10	192.168.10.0 /24	192.168.10.1	.10 to .200	.255
Faculty	20	192.168.20.0 /24	192.168.20.1	.10 to .200	.255
Student	30	192.168.30.0 /24	192.168.30.1	.10 to .200	.255
IT Department	40	192.168.40.0 /24	192.168.40.1	.10 to .100	.255
Guest	50	192.168.50.0 /24	192.168.50.1	.10 to .200	.255
Management (OOB)	99	10.0.0.0 /24	10.0.0.1	Static only	.255

5.4 Simulation Strategy

Cisco Packet Tracer 8.x serves as the simulation environment. The tool provides a sufficiently representative model of Cisco IOS behaviour to validate the design concepts central to this project, including VLAN trunking, Layer 3 routing, ACL enforcement, DHCP relay, and OSPF. Performance characteristics specific to hardware ASICs are outside the scope of this simulation.

To maintain simulation manageability while demonstrating architectural scalability, each access layer switch in the simulation represents the aggregate connectivity of an entire campus building. Multiple host PCs are connected to VLAN-assigned access ports on each switch. The following device set is used:

Device Role	Device Type	Qty	Represents
Core Layer Switch	Cisco Catalyst 3650 (L3)	2	Network backbone
Distribution Layer Switch	Cisco Catalyst 3650 (L3)	2	Policy enforcement tier
Access Layer Switch	Cisco Catalyst 2960 (L2)	5	One per campus building
DHCP and DNS Server	Server-PT	1	Centralised IP management
Internet Gateway Router	Cisco 2911 Router	1	WAN and NAT simulation
Host PCs (across all VLANs)	PC-PT and Laptop-PT	15	User endpoints per segment
Wireless Access Points	AccessPoint-PT	3	Student labs, Guest, Faculty areas

5.5 Configuration Phase

The configuration phase proceeds sequentially. Each sub-phase is validated before the subsequent sub-phase commences, ensuring that configuration errors are isolated and resolved before propagating into more complex fault conditions.

5.5.1 VLAN Configuration

VLANs are created on all distribution and access layer switches using consistent VLAN IDs and descriptive names as defined in the building mapping table. Access ports are assigned to their respective VLANs and the port mode is set to access to disable Dynamic Trunking Protocol negotiation. All inter-switch links are configured as 802.1Q trunk ports with an explicit allowed VLAN list restricted to only those VLANs whose traffic legitimately traverses each link. This selective trunk pruning reduces broadcast propagation scope and mitigates VLAN hopping exposure.

5.5.2 Inter-VLAN Routing

Inter-VLAN routing is implemented using Switched Virtual Interfaces on Layer 3 distribution switches. A dedicated SVI is created for each VLAN, with its assigned IP address serving as the default gateway for all hosts in that segment. This approach consolidates routing and policy enforcement at the distribution tier, consistent with Zero Trust boundary placement. The ip routing command is enabled on all distribution switches to activate Layer 3 forwarding. SVIs are assigned addresses from the gateway column of the IP addressing table and activated with the no shutdown command.

5.5.3 OSPF Dynamic Routing

OSPF Area 0 is configured on all Layer 3 devices, including core switches, distribution switches, and the internet gateway router. OSPF is selected over static routing because of its rapid convergence via the Dijkstra shortest path algorithm, its automatic route advertisement upon topology changes, and its suitability for multi-segment enterprise environments. Each VLAN subnet is advertised into OSPF Area 0 from the distribution switches using the network statement with appropriate wildcard masks. Passive interface configuration is applied to all SVI interfaces facing user-segment VLANs to suppress OSPF hello packets toward end-host segments, reducing protocol overhead and preventing unauthorised OSPF adjacency formation.

5.5.4 DHCP Configuration

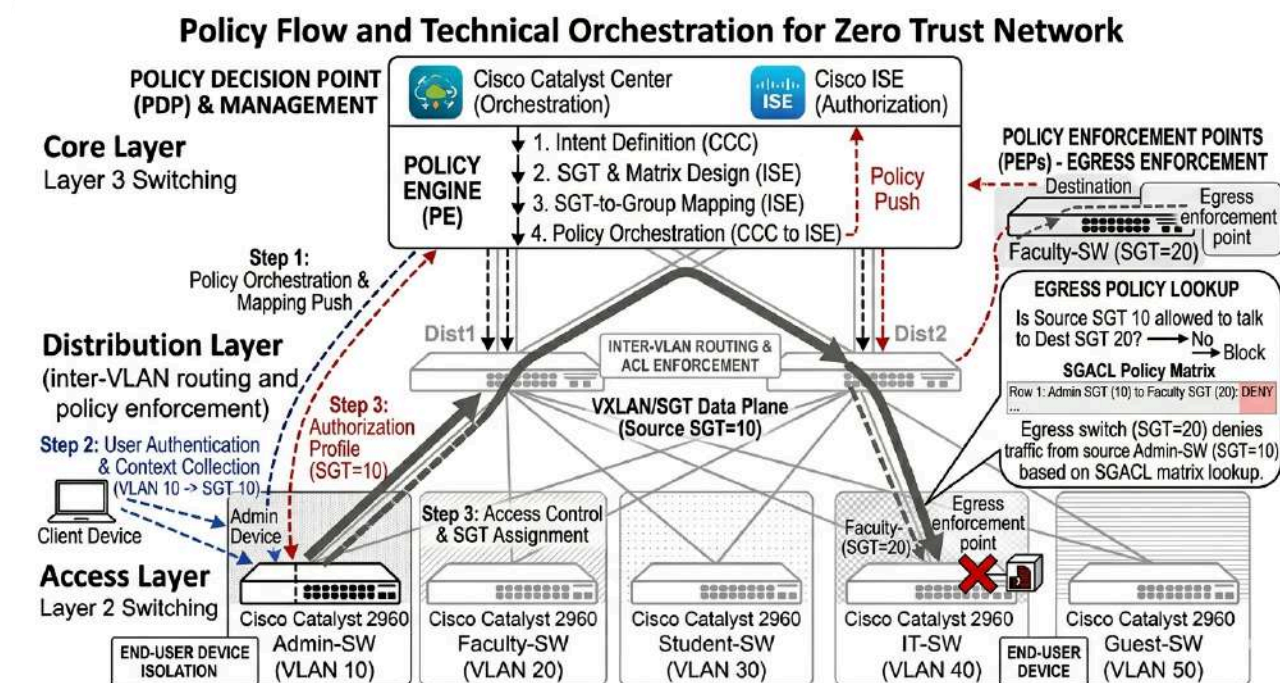
A centralised DHCP server is deployed in the IT Department VLAN and serves all five user VLANs. Because DHCP Discover broadcasts cannot cross VLAN boundaries without assistance, ip helper-address statements are configured on each VLAN's SVI at the distribution layer, relaying DHCP broadcasts as unicast packets to the server's IP address. This eliminates the operational overhead of maintaining per-VLAN DHCP servers while preserving centralised IP lifecycle management. Each DHCP scope defines the subnet, default gateway, DNS server address, and an exclusion range protecting statically assigned infrastructure addresses.

5.5.5 NAT Configuration

Dynamic NAT with Port Address Translation is configured on the internet gateway router to simulate controlled WAN access. An access list identifies the source subnets permitted to initiate outbound internet connections. Student, Faculty, Administration, and Guest subnets are included in the NAT source pool. The Guest VLAN, while permitted outbound internet access, is denied all inbound connectivity from internal segments by the GUEST-ISOLATE ACL applied at the distribution layer. A simulated ISP cloud in Packet Tracer represents the public internet, with the router's external interface assigned a routable public IP address.

5.5.6 Access Control List Design and Implementation

Extended Named ACLs constitute the primary Zero Trust enforcement mechanism in this design. Extended ACLs are used in preference to standard ACLs because they permit filtering based on both source and destination IP address as well as Layer 4 protocol and port number, enabling the granular role-based policy required by the security baseline. ACLs are applied inbound on SVI interfaces at the distribution layer, ensuring that policy is evaluated as traffic enters the routing domain before any forwarding decision is made.



The following table defines the complete ACL policy matrix governing inter-VLAN traffic:

Source VLAN	Destination VLAN	Action	Policy Rationale
Student (30)	Administration (10)	DENY all traffic	No student access to administrative systems
Student (30)	IT Department (40)	DENY all traffic	Prevent access to network infrastructure
Student (30)	Faculty (20)	DENY all traffic	Enforce academic boundary separation
Student (30)	Internet via NAT	PERMIT HTTP and HTTPS	Web-based academic resource access
Faculty (20)	Administration (10)	PERMIT selected TCP ports	Faculty ERP and administrative portal access
Faculty (20)	IT Department (40)	DENY SSH, SNMP, Telnet	Prevent unauthorised device management
Guest (50)	Any internal VLAN	DENY all internal traffic	Complete guest isolation: internet only
IT Department (40)	Any VLAN	PERMIT SSH, SNMP, ICMP	Supervisory network management access
Any VLAN	Any VLAN	DENY (implicit default)	Zero Trust default-deny baseline

Each ACL is assigned a descriptive name, including STUDENT-RESTRICT, GUEST-ISOLATE, and FACULTY-LIMIT, and is applied to the corresponding SVI using the ip access-group [NAME] in directive. The implicit deny any any statement at the end of each ACL enforces the Zero Trust default-deny posture without requiring an explicit terminal rule.

5.6 Testing and Validation

Validation is conducted in two sequential phases: connectivity verification and security enforcement verification. Both phases use systematic, documented test procedures with clearly defined expected and actual result comparisons. All results are recorded in a formal test report submitted as a project deliverable.

5.6.1 Connectivity Testing

Ping tests are conducted between representative host pairs within the same VLAN to confirm intra-segment reachability. Traceroute commands verify that inter-VLAN traffic traverses the intended routing path through the distribution tier. DHCP lease verification confirms that all hosts obtain correct IP addresses, subnet masks, and default gateway parameters. NAT functionality is verified by confirming that hosts in permitted source subnets successfully reach the simulated ISP address after address translation. The specific test cases are listed below.

- Intra-VLAN ping between two hosts in the Student VLAN. Expected result: success.
- Faculty PC to Administration portal server on a permitted TCP port. Expected result: success.
- All host PCs to their respective SVI default gateway. Expected result: success.
- DHCP lease verification across all five user VLANs. Expected result: correct subnet, gateway, and DNS parameters per scope.
- Student PC to simulated ISP address via NAT. Expected result: success with source address translated to public IP.

5.6.2 Security Enforcement Testing

ACL enforcement is verified by attempting connections that are defined as denied by the security policy matrix. Denied responses confirm that ACL rules are correctly applied and evaluated in the correct direction on the correct interface.

- Student PC ICMP ping to Administration PC in 192.168.10.0/24. Expected result: denied, destination unreachable.
- Student PC connection attempt to IT Department server in 192.168.40.0/24. Expected result: denied.
- Guest PC ICMP ping to Faculty workstation in 192.168.20.0/24. Expected result: denied.
- Guest PC connection to any internal subnet. Expected result: denied. Outbound NAT to internet remains successful.
- Faculty PC SSH attempt to distribution switch management IP in 10.0.0.0/24. Expected result: denied.
- IT Department PC SSH to access switch management IP. Expected result: success.

5.6.3 OSPF and Routing Verification

OSPF neighbour adjacency is confirmed on all Layer 3 devices. The routing table on each device is inspected to verify that all VLAN subnets appear as OSPF-learned routes. Convergence behaviour is tested by administratively disabling an inter-switch link and confirming that traffic reroutes through the alternate path within the OSPF convergence period.



5.7 Optimisation

5.7.1 Performance Improvements

- OSPF passive interface is applied to all access-facing SVIs to eliminate unnecessary OSPF hello traffic directed toward end-user segments and to prevent unauthorised OSPF adjacency formation.
- STP PortFast and BPDU Guard are enabled on all access ports to reduce end-device connection establishment time and to prevent rogue switch attachment on user-facing ports.
- Trunk link VLAN pruning restricts each trunk to carry only those VLANs whose traffic legitimately traverses that link, reducing unnecessary broadcast propagation.

5.7.2 Security Hardening

- VLAN 1 is disabled on all active ports. A non-default native VLAN (VLAN 99) is assigned to all trunk links to mitigate double-tagging VLAN hopping attacks.
- All unused switch ports are administratively shut down and assigned to a designated inactive VLAN (VLAN 999) to prevent connection of unauthorised devices.
- Port security is configured on critical access ports with a MAC address limit of one per port and violation mode set to restrict, which drops offending frames and generates a log event without disabling the port.
- DHCP snooping is enabled globally, with trusted ports defined only on uplinks toward the legitimate DHCP server, preventing rogue DHCP server injection attacks.

6. TOOLS AND TECHNOLOGIES

Tool or Technology	Specification	Rationale
Cisco Packet Tracer	Version 8.x	Industry-standard simulation tool with full support for required IOS features
Cisco Catalyst 3650	Layer 3 Switch (simulated)	Supports SVIs, OSPF, ACLs, and DHCP relay; suitable for core and distribution tiers
Cisco Catalyst 2960	Layer 2 Switch (simulated)	Access layer switch with full VLAN, STP, and port security support
Cisco 2911 Router	IOS Router (simulated)	Supports NAT, OSPF, and WAN interface simulation for the internet gateway function
OSPFv2	RFC 2328	Link-state protocol with rapid convergence; the de facto enterprise interior gateway protocol
IEEE 802.1Q	VLAN Trunking Standard	Industry standard for VLAN frame encapsulation on trunk links
Extended Named ACL	Cisco IOS ACL Framework	Provides source and destination-aware, Layer 4-capable access control for Zero Trust enforcement
DHCPv4	RFC 2131	Automates host IP address allocation across multiple VLAN scopes via relay agents
NAT with PAT	RFC 3022	Enables private-to-public address translation to simulate controlled WAN access
draw.io	Diagramming Tool	Production of network topology, campus layout, and ACL flow diagrams for documentation

7. EXPECTED OUTCOMES AND DELIVERABLES

7.1 Primary Deliverables

- A fully functional Cisco Packet Tracer simulation file containing the complete enterprise campus network, configured and validated against all stated objectives.
- A documented IP addressing and VLAN allocation table mapping all segments to physical buildings and defining all network parameters.
- A documented ACL ruleset with stated rationale for each rule, demonstrating alignment with the Zero Trust security policy baseline.
- This project proposal document, serving as the architectural blueprint and design rationale record.
- A formal test report documenting all connectivity and security test cases with expected and actual results, validating compliance with each stated objective.

7.2 Technical Outcomes

- Complete VLAN isolation verified between all five user segments, with no unauthorised inter-VLAN traffic observed during security testing.
- OSPF routing convergence achieved across all Layer 3 devices, with full routing table population confirmed and alternate path activation verified following simulated link failure.
- DHCP lease distribution confirmed for all user-facing VLANs with correct gateway and DNS parameters per scope.
- NAT functionality verified for all permitted source subnets; Guest VLAN confirmed as internet-accessible with no internal routing.
- All defined ACL deny rules confirmed to block traffic as specified; all defined permit rules confirmed to allow traffic as intended; implicit default deny confirmed at each SVI boundary.

7.3 Academic Outcomes

- Applied demonstration of enterprise network design principles including hierarchical architecture, VLAN segmentation, and dynamic routing.
- Practical proficiency in Cisco IOS configuration across Layer 2 and Layer 3 infrastructure devices.
- Demonstrated understanding of Zero Trust Security as a policy framework for access control in multi-segment network environments.

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8. CONCLUSION

This proposal presents a technically rigorous and professionally structured design for an enterprise campus network at UMT Lahore, built on the foundational principles of Zero Trust Security architecture. The design moves deliberately beyond the implicit trust model of perimeter-based networks by enforcing identity-independent, policy-driven access control at every segment boundary.

The three-tier hierarchical structure provides a scalable and maintainable framework. VLAN segmentation aligned with physical campus geography establishes clear and auditable trust boundaries. OSPF provides fault-tolerant dynamic routing. Extended ACLs enforced at the distribution tier implement the role-based access policy mandated by the Zero Trust baseline. DHCP and NAT complete the operational requirements for a functional enterprise environment.

The Cisco Packet Tracer simulation environment provides a controlled validation platform that faithfully reproduces the behavioural characteristics of production Cisco IOS infrastructure. The systematic testing methodology provides objective evidence that both functional requirements and security policy constraints are satisfied against each stated objective. The design is architecturally extensible to accommodate institutional growth without fundamental restructuring.

9. FUTURE WORK

The following extensions represent logical continuations of the current work and are identified for implementation in subsequent study, research, or professional practice.

- Integration of a RADIUS-based 802.1X port authentication framework to enforce identity-based network access at the physical port level, replacing VLAN membership as the sole trust boundary with cryptographically verified device and user credentials.
- Migration of the simulation to GNS3 or EVE-NG with production Cisco IOS images to eliminate behavioural approximations inherent in Packet Tracer and enable realistic performance benchmarking under simulated traffic loads.
- Deployment of a simulated Intrusion Detection System within the IT Department segment to provide traffic anomaly detection, complementing the preventive ACL controls with detective capability.
- Extension of the addressing scheme to dual-stack IPv4 and IPv6, incorporating OSPFv3 and DHCPv6 to reflect the long-term transition requirements of enterprise networks and the depletion of IPv4 public address space.
- Implementation of Software-Defined Networking principles using an OpenFlow-capable controller to centralise policy management and enable dynamic, context-aware access control that responds to real-time threat and posture data.
- Implementation of Quality of Service policies to differentiate traffic classes and prioritise latency-sensitive services such as voice and video over bulk data transfers, ensuring service quality for critical academic applications.

End of Project Proposal

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